Leblanc, Benjamin

Personal webpage
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EDUCATION

Doctor of Philosophy in Computer Science (Ph.D.), with honors (2022-2026)
Under the supervision of Prof. Pascal Germain
Laval University, Quebec

Certificate in Philosophy (2021-2023)
Laval University, Quebec

Master of Science in Computer Science, with thesis (M.Sc.) (2020-2021)
Under the supervision of Prof. Pascal Germain and Prof. François Laviolette
Laval University, Quebec

Bachelor of Science in Actuarial Science, international profile (B.Sc.) (2017-2020)
Laval University, Quebec

SCIENTIFIC ARTICLES

Publications at conferences and journals

- Bastian J., **Leblanc B.**, Germain P., Habrard A., Largeron C., Metzler G., Morvant E., and Viallard P. PAC-Bayesian Generalization Guarantees for Fairness on Stochastic and Deterministic Classifiers. **AISTATS (submitted; h5-index : 101)**, 2026. [preprint]

- Bastian J., **Leblanc B.**, Germain P., Habrard A., Metzler G., Morvant E., and Viallard P. On the Disintegration of the Stochastic Majority Vote: From PAC-Bayesian Bounds to a Self-Bounding Algorithm. **TMLR (submitted; h5-index : 50)**, 2026. [preprint]

- **Leblanc B.**, Bazinet M., D'Amours N., Drouin A., and Germain P. Generalization Bounds via Meta-Learned Model Representations: PAC-Bayes and Sample Compression Hypernetworks. **ICML (main track; h5-index : 272)**, 2025. PMLR 267:32830-32852. [publication][preprint]

- **Leblanc B.** and Germain P. Seeking interpretability and explainability in binary activated neural networks. **XAI World Conference (main track)**, 2024. Springer 2153:3-20. [publication][preprint]

- Sanchez-Reinoso Z., Bazinet M., **Leblanc B.**, Clément J.-P., Germain P., and Bazinet L. Application of machine learning tools to study the synergistic impact of physicochemical properties of peptides and filtration membranes on peptide migration during electrodialysis with filtration membranes. **Journal of Separation and Purification Technology (impact factor : 9.1)**, 2024. Elsevier 360:130877. [publication]

- Fortier-Dubois L., **Leblanc B**., Letarte G., Germain P., and Laviolette F. PAC-Bayesian Learning of Aggregated Binary Activated Neural Networks with Probabilities over Representations. **Canadian AI (main track)**, 2023. CAIAC 2023L18. [publication]

Publications at workshops

- **Leblanc B.**, Lebel L.-J., Kana T., and Kamel R. Simplicity Suffices for Parameter Noise Injection in Stochastic Gradient Descent. **IJCAI (Data Science Meets Optimisation (DSO) Workshop; h5-index : 136)**, 2026. [preprint]

- **Leblanc B.**, Bazinet M., D'Amours N., Drouin A., and Germain P. Sample Compression Hypernetworks: From Generalization Bounds to Meta-Learning. **NeurIPS (Neural Compression Workshop; h5-index : 371)**, 2024. [publication][preprint]

- **Leblanc B.** and Germain P. Seeking interpretability and explainability in binary activated neural networks. **IJCAI (Workshop on Explainable Artificial Intelligence (XAI); h5-index : 136)**, 2023. [publication]

- **Leblanc B.** and Germain P. A Greedy Algorithm For Building Compact Binary Activated Neural Networks. **Montreal AI Symposium**, 2022. [publication][preprint]

- Fortier-Dubois L., **Leblanc B.**, Letarte G., Germain P., and Laviolette F. Learning Aggregations of Binary Activated Neural Networks with Probabilities over Representations. **Montreal AI Symposium**, 2021. [publication]

Selected reports

- **Leblanc B.** and Germain P. A Framework for Bounding Deterministic Risk with PAC-Bayes: Applications to Majority Votes. ArXiv preprint, 2025. [preprint]

- **Leblanc B.** and Germain P. On the Relationship Between Interpretability and Explainability in Machine Learning. ArXiv preprint, 2023. [preprint]

GRANTS AND AWARDS

Doctoral thesis awarded honors
Laval University, 2026

François-Laviolette Scholarship for the development of responsible AI ($10,000)
Laval University, 2024

***Rouge et Or* academic star**
Rewarding the academic excellence of Laval University's student-athletes, 2024

***Rouge et Or* academic star**
Rewarding the academic excellence of Laval University's student-athletes, 2023

Best presentation award ($2,000)
NumériQC Week - AI and Data Rising Stars Competition, 2023

Pierre-Ardouin Prize
Rewarding the best session work, all graduate or multi-cycle courses combined, within the Department of Computer Science and Software Engineering at Laval University, 2022-2023

WORK EXPERIENCE

Machine Learning Researcher (Laval University) (2026-present)
Leading a machine learning research project on the use of language models to detect Parkinson's disease from speech samples, under the supervision of Prof. Cem Subakan.

University Instructor (Laval University) (2023-2024)
Taught the course Mathematics for Computer Scientists (MAT-1919) during the Fall 2023 and Winter 2024 sessions. Created exams, assignments, and quizzes. Course curriculum includes sets, relations, recurrences, trees, graphs, theory of finite and infinite sets.

Academic Tutor (Laval University) (2022-2025)
Provided one-on-one academic support to several students in mathematics and programming courses, helping them overcome conceptual difficulties and build problem-solving skills.

Business Intelligence Intern (Co-operators Insurance and Financial Services) (2019-2020)
Participated in the development and optimization of a tool for pricing in property and casualty insurance.

MENTORING

Undergraduate Research Supervision (2024-2026)
Led a research team of three undergraduate students (Lebel L.-J., Kana T., and Kamel R.) from project inception to their first academic publication in a peer-reviewed workshop:

- Leblanc B., Lebel L.-J., Kana T., and Kamel R. Simplicity Suffices for Parameter Noise Injection in Stochastic Gradient Descent. IJCAI (Data Science Meets Optimisation (DSO) Workshop; h5-index : 136), 2026. [preprint]

OUTREACH & COMMUNITY

Reviewer at conferences
NeurIPS (2023, 2026), ICML (2023, 2024, 2026), ICLR (2024, 2025), IJCAI (2024, 2025, 2027), CHI (2024), AISTATS (2026)

Reviewer at workshops
ICASSP Workshop XAI-SA (2024), NeurIPS workshop M3L (2024), ANNPR (2024), NLDL (2025), ICML workshop MOSS (2025)

Collaborator on the *I.A. Café* podcast
Contributed to several episodes covering AI vulgarization, current developments, and the ethical and social dimensions of artificial intelligence (2025-2026).

VP of the Graduate Student Association in Computer Science of Laval University
Organized social and cultural activities for graduate students in Computer Science at Laval University (2022-2023).

SPECIAL SKILLS

Languages
French (native), English (fluent), German (beginner)

Programming languages
Python (advanced), C++ (intermediate), R (intermediate), MiniZinc (beginner)